

IRIS — Conveyor Belt Digital Twin Inspection Report

Conveyor Unit: NMDC Kirandul Line 4B | Spec: ST-2500 (500m Loop) | Generated: 2026-09-28 19:31 UTC

Overall Health Score 99.9%	Remaining Useful Life 500 Hours	Advisory Level SCHEDULE REPLACEMENT	IS 11592 Safety Factor 7.3× (Pass)
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Plain-English Guide for Operators:

- Belt Health:** Operating at 99.9% health. Minor surface defects tracked. Safe for standard continuous ore conveying.
- Life Expectancy:** Estimated 500 hours (~20.8 days) before major structural reconditioning is recommended.
- Required Action:**

1. AI Vision Defect Ledger (Active Damage Events)

ID	Defect Class	Severity	Length	Confidence	RUL Impact
-	No defects detected	NONE	-	-	0.0 h

2. IS 11592 / DIN 22101 Engineering Compliance Audit

Engineering Parameter	Measured / Calculated	IS 11592 Limit	Compliance Status
Tight Side Tension (T1)	109,811 N	Max 320,000 N	PASS
Slack Side Tension (T2)	30,446 N	Min 12,000 N	PASS
Dynamic Safety Factor	7.30×	Min >= 8.0×	WARNING
Catenary Sag Ratio	1.31%	Max <= 2.0%	PASS
Drive Pulley Bearing Temp	35.4 °C	Max <= 80 °C	PASS
Vibration RMS Velocity	15.14 mm/s	ISO 10816 Limit	WARNING
Belt Speed & Position	3.15 m/s @ 3.1m	Rated 3.15 m/s	NORMAL

Autonomous AI Inspection Engineer: IRIS Digital Twin System (PS26008) Certified Digitally • IRIS-VAL-OK	Shift Maintenance In-Charge: NMDC Mining Division Signature: _____
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